From Master Integrals to Functions: the Short Version

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This is the compressed companion to the full lecture notes. Setting: the NNLO double-real channel of $pp \to h + X$; after reverse unitarity and IBP reduction there are **347 master integrals** depending on two Mandelstam ratios (v, w), $0 < v, w$, $v + w < 1$, in $d = 4 - 2\varepsilon$. The goal: every master as an explicit ε-expansion in iterated integrals, with exact constants.

1 Masters obey differential equations

Differentiating a master with respect to v or w produces integrals of the *same family* (same propagator set), which IBP reduces back to the masters of that family and its subsectors. Hence per family a closed first-order system

$$\mathrm{d}\vec{I} = A(v, w; \varepsilon)\,\vec{I}, \qquad A = A_v \mathrm{d}v + A_w \mathrm{d}w, \tag{1}$$

with rational A and the integrability (*flatness*) condition $\mathrm{d}A = A \wedge A$, certified exactly. Families were fixed at reduction time: *canonical families* `CF`n are equivalence classes of propagator sets under loop-momentum shifts ($GL(2, \mathbb{Z})$, cut lines to cut lines), detected via Pak's canonical Symanzik form and verified by the explicit shift. Differentiation is family-local, so the 347 masters split into **91 independent systems** (dimension 2–47). Ordered by sectors each system is *block lower triangular*: diagonal blocks couple one sector group's masters; off-diagonal blocks feed subsectors upward.

2 Blocks and the 173 classes

A *block* is a strongly connected group of masters (a diagonal block); the 91 systems contain 1119 of them. A *class* is an equivalence class of blocks under basis permutation, optionally composed with the crossing involution $v \leftrightarrow w$, as an exact matrix identity — there are only **173 classes**, because the same subtopologies recur inside many parent families. All hard analytic work is done once per class, and a registry maps each class solution back into every family that realizes it.

3 The ε-form and why it solves everything

A system is in *canonical (ε-)form* if

$$\mathrm{d}\vec{F} = \varepsilon \sum_a R_a \,\mathrm{d}\log \varphi_a(v, w)\,\vec{F}, \qquad R_a \text{ constant matrices}, \quad \varphi_a \ \varepsilon\text{-free } \textit{letters}. \tag{2}$$

Expanding $\vec{F} = \sum_k \varepsilon^k \vec{F}^{(k)}$ decouples the orders: $\mathrm{d}\vec{F}^{(k)} = \big(\sum_a R_a \mathrm{d}\log\varphi_a\big)\vec{F}^{(k-1)}$ — each order is a quadrature of the previous one. The solution is the path-ordered exponential, i.e. order ε^k

is a sum of k-fold *iterated integrals* (generalized polylogarithms, GPLs) over the alphabet, of uniform weight k. Example: letters $\{w, 1-w\}$ with nilpotent couplings generate $\log(1-w)$ at order ε and $\mathrm{Li}_2(w)$ at order ε^2. The ε-form thus converts coupled ε-entangled PDEs into explicit integrals; what remains is bookkeeping (paths, words) and boundary constants. Singularities are manifest: the connection's poles are exactly the letters' zeros, and the residue matrix R_a encodes the complete local behaviour there (exponents = $\varepsilon\times$ eigenvalues, logs from Jordan blocks).

4 Reaching the ε-form

Write the unknown basis change T: the connection transforms as $A' = T^{-1}AT - T^{-1}\mathrm{d}T$. Three stages:

1. **Diagonal blocks (per class).** Rational-ansatz solvers and Fuchsification/balance reductions (à la Lee) produce a candidate T; acceptance is only the exact reconstruction of the stored form from constant residues. Status: 173/173 classes done.

2. **Off-diagonal completion (per family).** With canonical diagonal blocks εe_k, complete one block row at a time by a unipotent gauge $T = 1 + D$ ($D^2 = 0$), which requires solving the *linear* strip equation

$$\mathrm{d}D_{kj} = \varepsilon(e_k D_{kj} - D_{kj} c_j) + \bar{b}_{kj} - \varepsilon \textstyle\sum_a K_a \mathrm{d}\log\varphi_a \tag{3}$$

   for a rational D_{kj} and constant residues K_a. Production method: ansatz over the letters with bounded numerator support, solved by finite-field sampling (exact linear algebra mod 31-bit primes), Chinese remaindering, rational reconstruction, held-out validation — and always a final *exact* identity as the certificate. The solution's gauge freedom is fixed toward the low-degree representative (pinning top numerator coefficients), which controls degree growth down the rows.

3. **Constant $T(\varepsilon)$.** Leftover ε-dependence in the residues stems from ε-dependent basis normalizations and is removed by a kinematics-independent $T(\varepsilon)$ found from sampled residue matrices and verified symbolically.

5 Roots, charts, Galois

Phase-space geometry injects square roots: the 3-particle boundary is the Källén polynomial $\lambda_1 = (1-v-w)^2 - 4vw$, and alphabets of many blocks contain $\sqrt{\lambda_1}$-type letters. The *number of roots* of a family = independent square classes among its radicands (44 families root-free, 31 one-root, 13 two-root, 3 three-root). A *chart* is a rational substitution making a root rational: e.g. $v = xy$, $w = (1-x)(1-y)$ gives $\lambda_1 = (x-y)^2$ (chart Kallen1); catalog charts Kallen1/2/3, Q4a/b, Bilinear115 cover the single roots, and iterated conic parametrization gives joint charts Kallen12/13/23 for the two-root families, whose ε-forms live in chart variables. For three roots no chart exists (the joint cover is a K3 surface); instead one computes in the multiquadratic *Galois algebra* $\mathbb{Q}(v,w,\varepsilon)[r_1,r_2,r_3]/(r_i^2 - q_i)$, whose $(\mathbb{Z}/2)^3$ sign group grades all functions into eight channels — linear algebra runs on the rational channel components, and the eight sign branches recombine as a certificate. This is the current frontier (`CF259/300/303`).

6 Transport

Solving in practice = choosing a base point and paths in the physical triangle, restricting letters to each segment (kept linear in the path parameter), and accumulating GPL words order by order. Flatness makes the result path-independent. The ε-depth per master is budgeted in advance from where it enters the assembled cross section (203 masters need the finite part only, 131 one pole deeper, 9 more); chart families transport in their chart variables.

7 Boundary constants

The general solution leaves one constant vector per family per ε-order. Almost none are new numbers; constraints are applied in order: (1) *volume rows* — the phase-space volume master is a closed-form Γ-ratio, seeding the lattice; (2) *inherited rows* — subsector masters already solved elsewhere constrain each family (families processed in dependency order); (3) *forbidden modes* — near $w \to 0$, $v \to 0$, and the soft surface $1 - v - w \to 0$ the integral *representation* allows only certain exponents $n + a\varepsilon$ (a power-counting statement per stratum, computed per chart), while the DE offers modes with a from residue eigenvalues; every representation-forbidden mode kills one constant; (4) *regularity* at letter zeros interior to an integral's physical region. The number of genuinely new constants per block is the *nullity* $\dim \ker C$ of the assembled constraint matrix, computed before evaluating anything; inventory-wide the surviving *boundary periods* are a short list of cut phase-space integrals at ordered limits, each established by an exact, numerics-free proof chain (with independent numerics as a guide only).

8 Endpoint behaviour: don't expand it away

Near an endpoint the solution is a sum of modes $u^{n+a\varepsilon}(\log u)^j$ with data read exactly off the residue matrices (this is "the singularity is in the matrix"). For the cross section one needs the distributional expansion

$$(1-w)^{-1+a\varepsilon} = \frac{1}{a\varepsilon}\,\delta(1-w) + \sum_{k\geq 0} \frac{(a\varepsilon)^k}{k!}\left[\frac{\log^k(1-w)}{1-w}\right]_+, \tag{4}$$

which exists only for the *unexpanded* power — expanding the master in ε first destroys the δ content. Masters therefore ship with exact indicial data alongside their GPL expansions, and endpoint extraction consumes the modes.

9 Certification

Every stage pairs a free-form search (ansatz, modular, numeric) with an exact certificate independent of the search: flatness identities, exact reconstruction of class forms, the exact family gauge identity (symbolically or by modular evaluation with held-out primes), and numerics-free proof chains for periods. The final artifact — 347 masters as certified ε-forms plus boundary data — verifies itself.

Pipeline map

Stage	Output	Count
IBP + canonical families	masters, family systems	347 masters, 91 systems
Blocking + classification	classes to solve once	1119 blocks $\to$ 173 classes
Class ε-forms	canonical diagonal blocks	173/173
Completion (strips + $T(\varepsilon)$)	family ε-forms	certified per family
Charts / Galois algebra	rational variables for roots	0/1/2 roots: charts; 3: algebra
Transport	GPL words, budgeted depth	per master
Boundary program	nullity $\to$ periods	few genuinely new periods
Endpoint modes	δ + plus distributions	from indicial data

Vocabulary in one breath. *Family*: propagator set mod momentum shifts (`CF`n). *Sector*: propagator subset. *Block*: strongly connected master group = diagonal block. *Class*: block mod permutation $\circ$ $(v \leftrightarrow w)$. *Connection*: A in $\mathrm{d}\vec{I} = A\vec{I}$. *$\varepsilon$-form*: $A = \varepsilon \sum R_a \mathrm{d}\log \varphi_a$, constant R_a. *Letter/alphabet*: the φ_a/their set. *Strip*: an off-diagonal block row being completed. *Root*: independent square class $\sqrt{q}$ in an alphabet. *Chart*: rational substitution rationalizing roots. *Transport*: building the path-ordered solution. *Period*: genuinely new boundary constant. *Nullity*: $\dim\ker(\text{constraints})$ = number of new periods. *Indicial data*: exponents/log powers at singular loci from residue matrices.